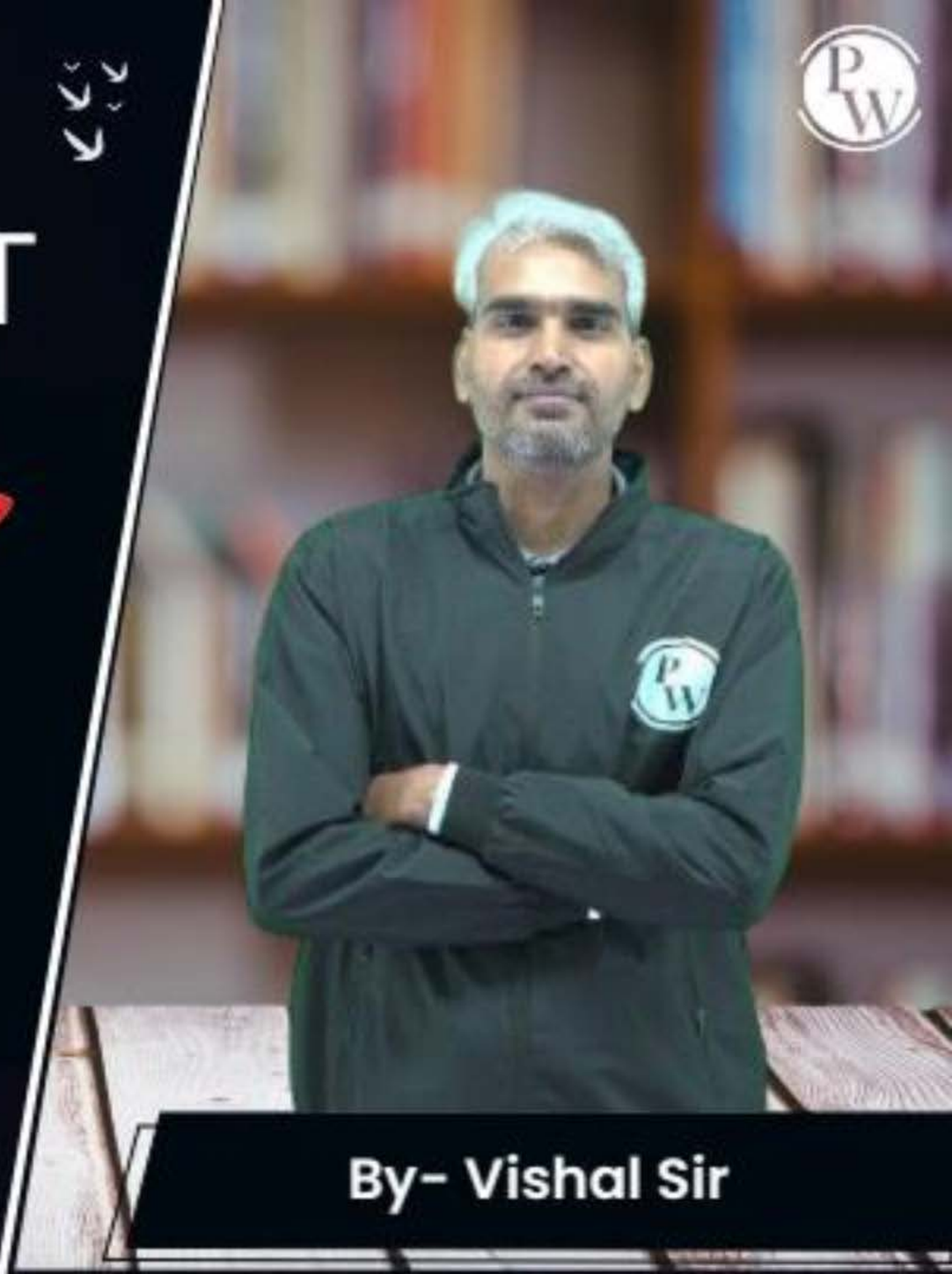
# Computer Science & IT



## Discrete Mathematics

## Mathematical Logic

**Lecture No. 07**



By- Vishal Sir

# Recap of Previous Lecture



Topic — Negation of statement formula

Topic — Sentences with multiple quantifiers

Topic — Relationship diagram

Slide

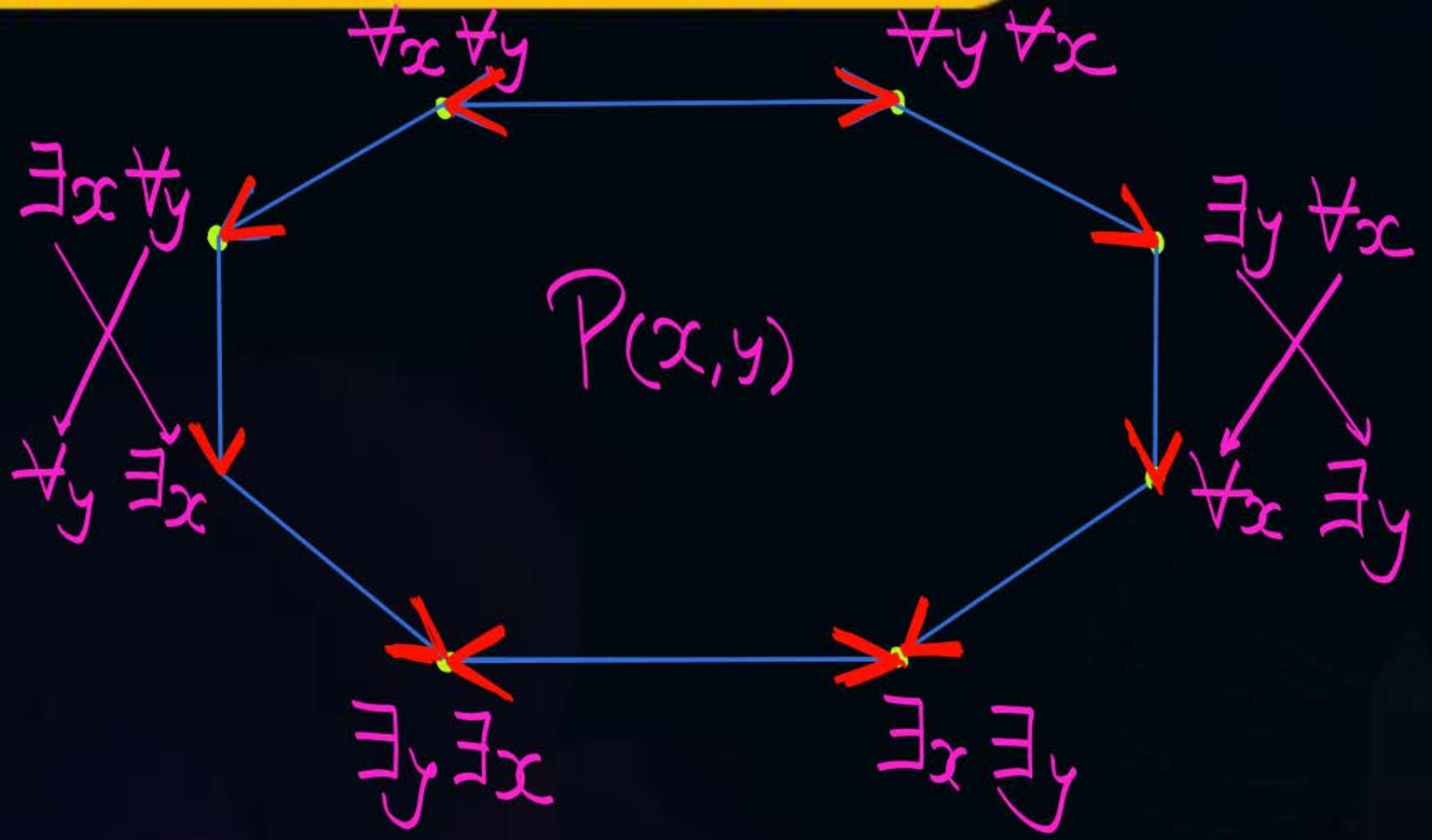
# Topics to be Covered



- Topic: Rules of inferences with respect to predicate logic
- Topic: Important logical implications and equivalences
- Topic: Practice questions

Slide

# Topic : Relationship diagram




$\longleftrightarrow$ represent Equivalence

&

$\longrightarrow$ represent logical implication



# Topic : Rules of Inferences w.r.t. predicate logic

- In predicate logic, we can use some additional inference rules, along with all the rules of inference we have discussed in propositional logic.

- Additional Rules of Inference w.r.t. Predicate Logic

1. Universal Instantiation (Universal Specification)
2. Universal Generalization
3. Existential Instantiation (Existential Specification)
4. Existential Generalization



# Topic : Universal Instantiation

## Universal specification

$$\frac{\forall_x \{P(x)\}}{\therefore\ P(a)} \text{ is valid}$$

$\therefore$ $P(a)$ is true for any 'a' in the domain

for all (or) 'a' in the domain



# Topic : Universal Generalization

P(a) is true for all 'a' in domain

$\therefore \forall_x \{P(x)\}$

We can use any variable for generalization

{ i.e. $\forall_z \{P(z)\}$ is also Valid }



# Topic : Existential Instantiation

$$\frac{\exists_x \{P(x)\}}{\therefore P(a) \text{ is true for some 'a' in domain}} \quad \text{is Valid}$$



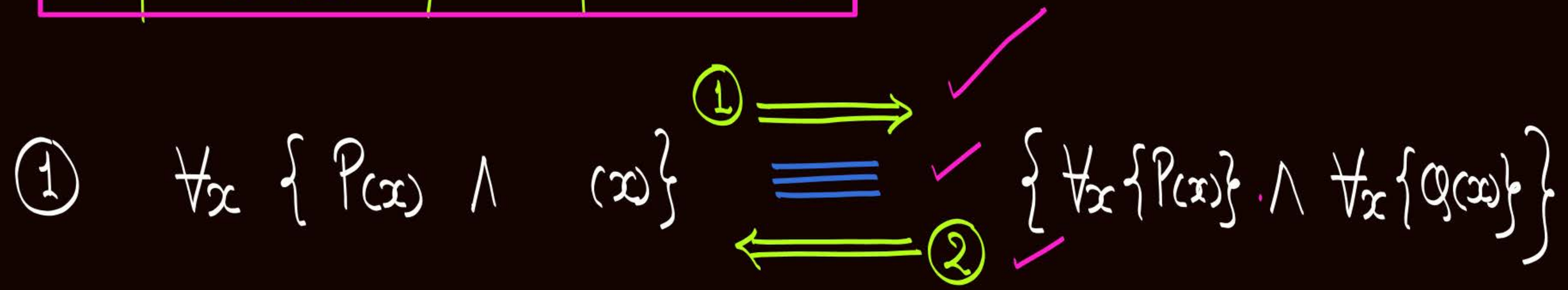
# Topic : Existential Generalization

P(a) is true for some (at least one) 'a' in domain

$\therefore \exists_z \{P(z)\}$

$\forall x \{P(x)\}$ will be invalid

# Equivalences & Implications



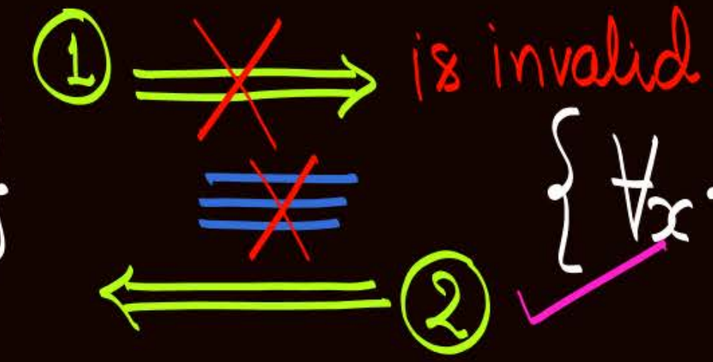
# Equivalences & Implications

② $\exists_x \{ P(x) \vee Q(x) \}$ $\equiv$ $\{ \exists_x \{P(x)\} \vee \exists_x \{Q(x)\} \}$

① ⟹ ✓

② ⟸ ✓

# Equivalences & Implications




③ $\forall x \{ P(x) \vee Q(x) \}$ $\{ \forall x \{P(x)\} \vee \forall x \{Q(x)\} \}$

$U = \{a, b\}$

$P(a) = T \vee Q(a) = F$
$P(b) = F \vee Q(b) = T$

$T \rightarrow T$ then

LHS = True

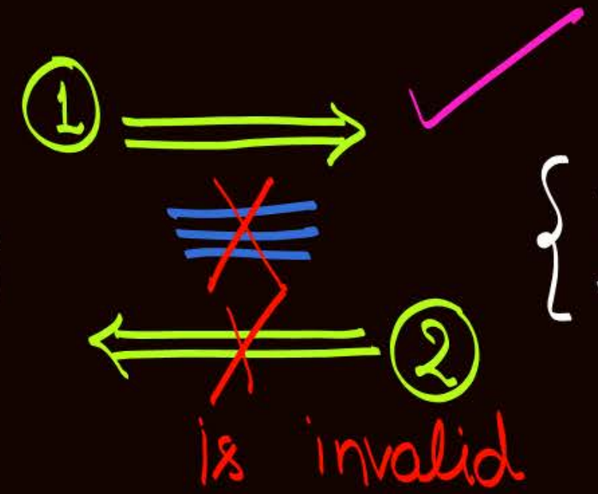
$P(a) = T$ $Q(a) = F$
$P(b) = F$ $Q(b) = T$

$\forall x \{P(x)\} = F$ $\vee$ $\forall x \{Q(x)\} = F$

$\vee$ → False

∴ RHS = False

# Equivalences & Implications

④ $\exists_x \{ P(x) \wedge Q(x) \}$  $\{ \exists_x \{P(x)\} \wedge \exists_x \{Q(x)\} \}$



# Topic : Equivalences & Implications

6

$\forall x \{P(x) \leftrightarrow Q(x)\} \Rightarrow (\forall x \{P(x)\} \leftrightarrow \forall x \{Q(x)\})$

1. $\forall x\,[P(x) \wedge Q(x)] \equiv [\forall x\, P(x)] \wedge [\forall x\, Q(x)]$

2. $\exists\, x\,[P(x) \vee Q(x)] \equiv [\exists\, x\, P(x)] \vee [\exists\, x\, Q(x)]$

3. $[\forall x\, P(x)] \vee [\forall x\, Q(x)] \Rightarrow \forall x\,[P(x) \vee Q(x)]$

4. $\exists\, x\,[P(x) \wedge Q(x)] \Rightarrow [\exists\, x\, P(x)] \wedge [\exists\, x\, Q(x)]$

5. $\forall x\,[P(x) \rightarrow Q(x)] \Rightarrow [\forall x\, P(x)] \rightarrow [\forall x\, Q(x)]$



# Topic : Some important equivalences

1. $\forall x\,[P(x) \wedge Q] \equiv \forall x\,P(x) \wedge Q$

2. $\exists x\,[P(x) \vee Q] \equiv \exists x\,P(x) \vee Q$

3. $\forall x\,[P(x) \vee Q] \equiv \forall x\,P(x) \vee Q$

4. $\exists x\,[P(x) \wedge Q] \equiv \exists x\,P(x) \wedge Q$

Note: In the given formulae 'Q' is a predicate formula without any free occurance of 'x'

⇓

i.e. Truth value of Q does not vary with 'x'

⇓

i.e. if Q is true, then it will remain true for all x

if Q is false then it will remain false for all x



# Topic : Some important equivalences

1. $\forall x\,[P(x) \wedge Q] \equiv \forall x\,P(x) \wedge Q$

2. $\exists x\,[P(x) \vee Q] \equiv \exists x\,P(x) \vee Q$

3. $\forall x\,[P(x) \vee Q] \equiv \forall x\,P(x) \vee Q$

4. $\exists x\,[P(x) \wedge Q] \equiv \exists x\,P(x) \wedge Q$

Case ① Q = False

When Q is false, the both LHS and R.H.S. will always return false for all values of x.

i.e. When Q = False, then same truth table for LHS & RHS

Case ② Q = True

① For LHS to be true, 'P' must be true for all x, and in that case $\forall x\{P(x)\}$ will also return true, Hence RHS will also be true

② For RHS to be true $\forall x\{P(x)\}$ must be true, Hence for all 'x' $P(x) \wedge Q$ will also be true, Hence LHS will also be true



# Topic : Some important equivalences

1. $\forall x\,[P(x) \wedge Q] \equiv \forall x\,P(x) \wedge Q$

2. $\exists x\,[P(x) \vee Q] \equiv \exists x\,P(x) \vee Q$

3. $\forall x\,[P(x) \vee Q] \equiv \forall x\,P(x) \vee Q$

4. $\exists x\,[P(x) \wedge Q] \equiv \exists x\,P(x) \wedge Q$

H.W. Check for validity of all the equivalences.

Hint: Check w.r.t. two cases

(i) $Q$ = True

(ii) $Q$ = False



# Topic : Some important equivalences

5. $\forall x\,[P \to Q(x)] \equiv P \to \forall x\,Q(x)$

6. $\exists x\,[P \to Q(x)] \equiv P \to \exists x\,Q(x)$

In first two predicate formula 'P' is a predicate formula without any free occurance of 'x'.

7. $\forall x\,[P(x) \to Q] \equiv \exists x\,P(x) \to Q$

8. $\exists x\,[P(x) \to Q] \equiv \forall x\,P(x) \to Q$

In 7th and 8th predicate formula 'Q' is the predicate formula without any free occurance of x.



# Topic : Some important equivalences

5. $\forall x\,[P \to Q(x)] \equiv P \to \forall x\, Q(x)$

$\forall x\{\sim P \vee Q(x)\} \equiv \sim P \vee \forall x\{Q(x)\} \equiv P \to \forall x\{Q(x)\} \equiv$ R.H.S.

6. $\exists x\,[P \to Q(x)] \equiv P \to \exists x\, Q(x)$

$\exists x\{\sim P \vee Q(x)\} \equiv \sim P \vee \exists x\{Q(x)\} \equiv P \to \exists x\{Q(x)\} \equiv$ RHS.

7. $\forall x\,[P(x) \to Q] \equiv \exists x\, P(x) \to Q$

$\equiv \forall x\{\sim P(x) \vee Q\} \equiv \forall x\{\sim P(x)\} \vee Q$

$\sim(\exists x\{P(x)\}) \vee Q \equiv \exists x\{P(x)\} \to Q \equiv$ RHS.

8. $\exists x\,[P(x) \to Q] \equiv \forall x\, P(x) \to Q$

$\exists x\{\sim P(x) \vee Q\} \equiv \exists x\{\sim P(x)\} \vee Q \equiv \sim(\forall x(P(x))) \vee Q \equiv \forall x\{P(x)\} \to Q \equiv$ RHS

$\sim \forall x\{P(x)\}$



# Topic : Some important equivalences

5. $\forall x\,[P \to Q(x)] \equiv P \to \forall x\, Q(x)$

6. $\exists x\,[P \to Q(x)] \equiv P \to \exists x\, Q(x)$

7. $\forall x\,[P(x) \to Q] \equiv \exists x\, P(x) \to Q$

8. $\exists x\,[P(x) \to Q] \equiv \forall x\, P(x) \to Q$

## [MSQ]

#Q. Let $P(x)$ and $Q(x)$ be arbitrary predicates. Which of the following statements is always TRUE?

A $(\forall x(P(x) \vee Q(x))) \Rightarrow ((\forall x P(x)) \vee (\forall x Q(x)))$

B $(\forall x(P(x) \Rightarrow Q(x))) \Rightarrow ((\forall x P(x)) \Rightarrow (\forall x Q(x)))$

C $(\forall x(P(x) \Rightarrow \forall x Q(x))) \Rightarrow (\forall x\ (P(x) \Rightarrow Q(x)))$

D $((\forall x(P(x)) \Leftrightarrow (\forall x Q(x))) \Rightarrow (\forall x(P(x) \Leftrightarrow Q(x)))$

[MSQ]

#Q. Which of the following predicate calculus statements is/are valid

A $(\forall x)P(x) \vee (\forall x)Q(x) \rightarrow (\forall x)\{P(x) \vee Q(x)\}$

B $(\exists x)P(x) \wedge (\exists x)Q(x) \rightarrow (\exists x)\{P(x) \wedge Q(x)\}$

C $(\exists x)\{P(x) \vee Q(x)\} \rightarrow (\forall x)P(x) \vee (\forall x)Q(x)$

{it is valid only if domain is not empty}

D $(\exists x)\{P(x) \vee Q(x)\} \rightarrow \{\exists x\, P(x) \vee \forall x\, Q(x)\}$

is invalid

if P = false for all, and Q is true for some but not for all, then LHS = True but RHS = False

[MSQ]

#Q. Which of the following predicate calculus statements is/are valid

A $\forall(x)\big(P(x) \Rightarrow Q(x)\big) \Rightarrow \big(\big(\forall(x)P(x)\big) \Rightarrow \big(\forall(x)Q(x)\big)\big)$

B $\exists x\big(P(x) \vee Q(x)\big) \Rightarrow \big(\big(\exists xP(x)\big) \Rightarrow \big(\exists xQ(x)\big)\big)$

T → F = T

if P = True for some but Q = False for all, then LHS = True & RHS = False

C $\exists x\big(P(x) \wedge Q(x)\big) \Leftrightarrow \big(\big(\exists xP(x)\big) \wedge \big(\exists xQ(x)\big)\big)$

D $\forall(x)\exists y\, P(x,y) \Rightarrow \exists y\, \forall(x)P(x,y)$

[MSQ]

#Q. Which of the following is a valid first order formula? (Here $\alpha$ and $\beta$ are first order formulae with $x$ as their only free variable)

let $\alpha \equiv P(x)$, $\beta = Q(x)$

A $((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]) \Rightarrow (\forall x)[\alpha \Rightarrow \beta]$

$\forall x\{P(x)\} \Rightarrow \forall x\{Q(x)\} \not\Rightarrow \forall x\{P(x) \rightarrow Q(x)\}$

B $((\forall x)[\alpha] \Rightarrow (\exists x)[\alpha \wedge \beta])$

$T \rightarrow F \equiv F$

if $\alpha$ = True for all, but $\beta$ = false for all then "$\rightarrow$" will return false

C $((\forall x)[\alpha \vee \beta] \Rightarrow (\exists x)[\alpha]) \Rightarrow (\forall x)[\alpha]$

$\phi \rightarrow T$, $T \rightarrow T$

let $\forall x(\alpha)$ = false

$F \rightarrow$ false

{ let $\alpha$ = false for some but not for all }

D $((\forall x)[\alpha \Rightarrow \beta] \Rightarrow ((\forall x)[\alpha]) \Rightarrow (\forall x)[\beta])$

$\forall x\{P(x) \rightarrow Q(x)\} \Rightarrow (\forall x\{P(x)\} \rightarrow \forall x\{Q(x)\})$

[MSQ]

#Q. Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first-order formula with x as a free variable and $\beta$ is a first-order formula with no free variable.

H.W.

**A** $[\beta \rightarrow (\exists x, \alpha(x))] \rightarrow [\forall x, \beta \rightarrow \alpha(x)]$

**B** $[\exists x, \beta \rightarrow (\alpha(x))] \rightarrow [\beta \rightarrow (\forall x, \alpha(x))]$

**C** $[(\exists x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$

**D** $[(\forall x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$

[MSQ]

H.W.

#Q. Which one of the following predicate formulae is NOT logically valid? Note that $W$ is a predicate formula without any free occurrence of $x$.

A $\exists x(p(x) \wedge W) \equiv (\exists x\, p(x)) \wedge W$

B $\forall x(p(x) \rightarrow W) \equiv (\forall x\, p(x)) \rightarrow W$

C $\exists x(p(x) \rightarrow W) \equiv (\forall x\, p(x)) \rightarrow W$

D $\forall x(p(x) \vee W) \equiv (\forall x\, p(x)) \vee W$

[MSQ]

#Q. What is the correct translation of the following statement into mathematical logic?

"some real numbers are rational"

A $\exists x\ (real(x) \vee rational(x))$

B $\exists x\ (real(x) \wedge rational(x))$

C $\forall x\ (real(x) \rightarrow rational(x))$

D $\forall x\ (rational(x) \rightarrow real(x))$

[MSQ]

#Q. What is the logical translation of the following statement?

"None of my friends are perfect"

A $\exists x\,(F(x) \wedge \sim P(x))$

B $\exists x\,(\sim F(x) \wedge \sim P(x))$

C $\exists x\,(\sim F(x) \wedge P(x))$

D $\sim (\exists x\,(F(x) \wedge P(x)))$

H.W.

**[MSQ]**

#Q. The CORRECT formula of the sentence "not all rainy days are cold"

**A** $\forall\, d\ (rainy(d)\ \wedge \sim cold(d))$

**B** $\exists\, d\ (\sim\ rainy(d)\ \rightarrow\ cold(d))$

**C** $\forall\, d\ (\sim\ rainy(d)\ \rightarrow cold(d))$

**D** $\exists\, d\ (rainy(d)\ \wedge \sim\ cold(d))$

[MSQ]

H.W.

#Q. Which one the following is the most appropriate logical formula to represent the statement "gold and silver ornaments are precious". The following notations are used: $G(x)$:$x$ is a gold ornament

$S(x)$:$x$ is a silver ornament $P(x)$:$x$ is precious

A $\forall x(P(x) \rightarrow G(x) \wedge S(x))$

B $\exists x\,((G(x) \wedge S(x)) \rightarrow P(x))$

C $\forall x\,((G(x) \wedge S(x)) \rightarrow P(x))$

D $\forall x\,((G(x) \vee S(x)) \rightarrow P(x))$

H.W.

[MSQ]

#Q. Which one of the first order predicate calculus statements given below correctly expresses the following English statements? "Tigers and lions attack if they are hungry or threatened".

A $\forall x[(tiger(x) \wedge lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \rightarrow attacks(x))]$

B $\forall x[(tiger(x) \vee lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \wedge attacks(x))]$

C $\forall x[(tiger(x) \wedge lion(x)) \rightarrow (attacks(x) \rightarrow (hungry(x) \vee threatened(x)))]$

D $\forall x[(tiger(x) \vee lion(x)) \rightarrow ((hungry(x) \vee threatened(x)) \rightarrow attacks(x))]$



# 2 mins Summary

- Topic: Rules of inferences with respect to predicate logic
- Topic: Important logical implications and equivalences
- Topic: Practice questions

Slide

# THANK - YOU